Social Trust as an Observational Sensor for Institutional Niche Dynamics

Somatic Unsafety, Behavioural Hedging, Schema Lag and Cross-Regime Telemetry in DST–NCT

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Abstract

Social trust is frequently represented either as a generalised attitude or as an immaterial stock that reduces the cost of institutional coordination. Neither representation is adequate for identifying divergence between the declared continuity of a long-lived institution and the physical reliability of the human niche on which it depends. This paper formalises trust as a domain-specific, directional observational sensor operating on three response times: autonomic detection ($t_{autonomic}$), pragmatic reliance and verification (t_{sig}), and inherited declarative schemas (t_{schema}). It preserves the Dissipative Structure Theory–Niche Construction Theory (DST–NCT) state-space $x(t) = (b(t), m(t), a(t), u(t))$, without treating trust as a fifth fundamental state or an exergy stock. A history-dependent observation operator links the three clocks to physical exposure, care provisioning, network relations and institutional information. The model derives conditional somatic suppression costs, a recovery-dependent allostatic burden, heterogeneous public-recognition thresholds and hysteretic coordination transitions. It explicitly separates these informational transitions from physical separatrix crossings, relational trust from coercive assurance, and confidence from actual buffer coverage. A hardened five-domain telemetry matrix addresses administrative gaming, denominator changes and physiological confounding. A comparative test at $t_0 = 2026$ uses Sweden, the United Kingdom and the United States to distinguish schema-locked divergence, externally coupled reliance and heterogeneous modular buffering. The country portraits are conditional structural interpretations anchored to identified observations, not measured national three-clock scores. Falsification conditions and an empirical research design specify what evidence would establish discriminatory power beyond existing trust surveys and macroeconomic accounting.

Keywords: *DST–NCT; relational trust; allostasis; generalized unsafety; information asymmetry; institutional resilience; hysteresis; care networks; network thresholds; resource buffering.*

1 Problem, scope and causal architecture

The foundational DST–NCT model describes biological activity and population structure b(t), the material niche m(t), inherited institutional rules and claims a(t), and external physical driving conditions u(t). The human institutional application permits $a(t) = (a_{rules}(t), a_{claims}(t), a_{models}(t), \ldots)$, while m(t) includes stored usable exergy B(t) and persistent functional structures s(t). The core material balances remain $dB/dt = P_{in} - P_{out} - P_{dest}$ and $ds/dt = G_s + R_s - D_{s,}$ with all quantities defined over a specified system boundary. The equations and physical constraints are not modified by the present paper [1].

The governing distinction is between a claim, an observation and the physical event required to realise that claim. A hospital budget, an administrative declaration of service continuity, a household's expectation of safe care and a delivered episode of physical care are different kinds

of object. They can interact causally, but they cannot be converted into one another by relabelling. Accordingly, observed trust does not constitute stored work potential, a reserve of personnel-hours or functioning infrastructure. Reduced coordination friction can conserve resources; it cannot supply missing resources except through an independently measured change in labour, throughput, maintenance, access or allocation [1, 2].

Conventional interpersonal trust surveys, institutional-confidence instruments and national-accounting indicators observe different compressed features of this coupled system. A survey can correctly measure a stated attitude while remaining silent about how much checking a household performs before entrusting a dependent relative to care, what extra labour a frontline team expends on defensive documentation, or whether a supplier can physically deliver goods by the necessary date. The present model therefore investigates whether discordance among physiological response, demonstrated practical reliance and inherited declarative expectation improves observation of hidden transitions.

The contribution is conditional, not diagnostic by assertion. Trust-related behaviour can change without structural cannibalisation: acute local hazard, sudden information, class-specific exposure, efficient substitution and deliberate contractual redesign can all alter observed reliance. Likewise, structural cannibalisation can develop with little publicly visible distrust under coercion, constrained exit, physiological adaptation or selection of the remaining population. The model must discriminate such cases rather than assume the proposed mechanism in advance.

1.1 Methodological lock on causal ordering

When sustained reserve depletion or structural under-maintenance is independently established, Chapter 12 of the parent framework orders horizon compression and centripetal fiduciary defection downstream of material niche contraction; they can accelerate subsequent deterioration but are not retroactively its initiating energetic cause [1]. The conditional chain is: material and temporal constraints → biological and pragmatic recalibration → institutional or network rerouting → changed rates of maintenance, extraction and coordination. In a different regime, failed information or interpersonal betrayal can occur without negative dB/dt. These cases must not be classified as structural cannibalisation merely because trust observables deteriorate.

The non-substitution rule is strict. A specified reserve covers a constant shortfall over $T_{cover} = B_{available}/P_{shortfall}$ only if adequate delivery power exists. Repeated withdrawals require assessment of T_{refill} relative to T_{recur}; if $T_{refill} > T_{recur}$ and neither replenishment nor withdrawal changes, progressive depletion follows. Persistent infrastructure deteriorates when $D_s > G_s + R_s$ for the selected structural service. A change in $C_{i,}$ survey sentiment or administrative confidence cannot reverse either physical inequality without corresponding measured physical flows. $T_{cover} > T_{recur,}$ sometimes used as a general shorthand, is not a sufficient repeated-mobilisation test: coverage duration, recurrence, replenishment and power are distinct [1].

1.2 Unit and notation discipline

Time t is physical elapsed time. The terms $t_{autonomic,}$ t_{sig} and t_{schema} refer to characteristic response times, not independent temporal axes. The parent framework uses H for a representational or forward-commitment horizon; in this paper $H_{i,d}$ is the actor's practical commitment horizon for domain d, H_{inst} is the institution's representational horizon, and H_{ext} is the relevant external counterparty's horizon. These must not be silently equated. All observation indices below are dimensionless unless otherwise stated. P_{vig} and P_{sup} are energetic power rates in watts only when calibrated from actual expenditure. ΦS is the core entropy-transfer rate in J/(K·s), not another name for verification expenditure [1].

2 Three-clock trust telemetry

Three layers are distinguished by mechanism, latency, observable and confounding conditions. They are neither a universal sequence nor a single hierarchy of confidence. Acute bodily arousal can be strong while social reliance remains rational under high stakes; conversely calm bodily behaviour can coexist with calculated refusal to enter a contract.

Layer 1 — autonomic unsafety ($t_{autonomic}$): short-latency shifts in physiological activation, vigilance, sensory monitoring and recovery under uncertain safety cues. Generalized Unsafety Theory of Stress (GUTS) provides a candidate mechanism whereby insufficient safety information sustains defensive physiological readiness [3]. No unique polyvagal pathway is required. Visible motor stiffness, self-touch, startling, screen use and chemosensory cues are non-specific and may reflect heat, exertion, health, habit or sensory preference [4, 5].

Layer 2 — pragmatic reliance and signalling (t_{sig}): the observed allocation of vulnerability without continuous independent verification, collateral or coercive assurance, measured across a defined counterparty and duration. The construct is directional and domain-specific. It is inferred from choices only where actors possess a materially meaningful set of alternatives; compulsory continued participation is not a measure of voluntary reliance [6, 7].

Layer 3 — declarative continuity schema (t_{schema}): inherited cognitive expectations, legitimate procedural normality and publicly expressible beliefs about social reliability and succession. Schemas are shaped by developmental history and repetition but can update; public discourse can switch rapidly even while an internal developmental template persists. Survey questions observe a mixture of these features and current politics, and cannot be assigned uniquely to a pure schema without further measurement [8].

Table 1. Three-clock architecture

Layer	Typical characteristic response	Primary readout	Principal limitation
1. Autonomic	Milliseconds to minutes in acute sampling; recovery across hours or days	Within-person deviations in arousal, vigilance and recovery	Physiological non-specificity; context-dependent calibration
2. Pragmatic	Episodic decisions to months or years	Real unverified commitment, checking, withdrawal, local substitution	Changes in incentives, alternatives and domain exposure
3. Declarative	Developmental learning over years; public expression can change faster	Stated reliability and normality expectations	Identity scripts, reporting selection and different question wording

The clocks describe response tendencies rather than invariant chronological bounds. A new, unambiguous failure can instantly alter Layer 2 behaviour; repeated provision can slowly restore bodily safety; a political event can rapidly change a public answer without erasing earlier learning.

3 Directional observation operator and cross-layer coupling

For observer i, counterparty j, vulnerability domain d and horizon H, define the directional operator:

$$T_L(i \to j \mid d, H, t) = O_L(i, j, d, H, x[t-w,t], N(t), e_i(t)) + \varepsilon_L(t)$$

L ∈ {autonomic, pragmatic, schema}; N(t) is the relevant topology of support and observation relations already represented within b(t) and a(t); $e_i(t)$ specifies exposure, practical alternatives, previous experience and measurement context; and ε_L is observation error. No conserved trust stock and no additional core state equation result. $T_L(i \to j)$ generally differs from $T_L(j \to i)$. A high value in d = data accuracy does not establish reliable physical care in d = eldercare. Nor does an

average across counterparty pairs preserve information about network modularity or the locations of bottlenecks [7, 9].

For dynamic analysis let A_i be normalised somatic unsafety, V_i normalised verification or hedging, and C_i normalised declarative continuity expectation, each within [0,1]. High A_i and V_i represent increased defence; high C_i indicates continued declared continuity. They are observational coordinates, not fundamental physical stocks. One parsimonious filter system is:

$$t_{autonomic} \cdot dA_i/dt = F_A(q_i, r_i, z_i) - A_i$$

$$t_{sig} \cdot dV_i/dt = F_V(A_i, f_i, n_i, c_i) - V_i$$

$$t_{schema} \cdot dC_i/dt = F_C(h_i, p_i, o_i) - C_i$$

The bounded response functions F_A, F_V and F_C are estimated rather than assumed linear. Here q_i is experienced unpredictability, r_i recoverable safety, z_i physiological calibration, f_i verified practical failures, n_i observed network behaviour, c_i checking and substitution costs, h_i developmental expectation, p_i repeated procedure outcomes and o_i continuity signals. The equations encode lagged observation of x(t), not independent trust dynamics free of biological or institutional implementation.

The induced feedback has five legs. Material and care conditions alter q_i and f_i. Persistent bodily unsafety changes the salience of checking and defensive routing. Checking and rerouting change labour allocation and logistical coordination within b(t) and a(t). Institutional declarations and media can sustain C_i without changing q_i or f_i. Finally, expressive rules can shape how bodily activation is managed and communicated, modifying recovery and the visibility of the other feedbacks. The sign of each coupling remains conditional on the observed domain and material options.

3.1 Schema–substrate divergence

Define the lower-layer cooperation index $R_i = w_A(1 - A_i) + w_V(1 - V_i)$, with $w_A \geq 0$, $w_V \geq 0$ and $w_A + w_V = 1$. Define $D_i = C_i - R_i$ and $D_{ipositive} = \max(0, D_i)$. Positive D_i represents declared continuity exceeding measured lower-layer cooperation. Negative D_i represents practical or bodily reliability exceeding declarative expectations. Neither sign identifies dB/dt or D_s by itself.

The relevant aggregate is the distribution of D_i across exposure, care need, class, region, organisation and network position, not an unweighted national mean. Layer 1 and Layer 2 may diverge from each other; therefore, w_A and w_V should be fitted by domain and sensitivity-tested, not chosen to manufacture a predicted national ranking. The preferred reporting format includes separate A_i, V_i and C_i estimates before any composite D_i.

4 Somatic Double-Bind, allostatic accumulation and horizon compression

A somatic double-bind exists where direct exposure sustains unsafety, while expressive or organisational rules require suppression of ordinary defensive signalling and where physical exit or recovery is constrained. Strict conflict avoidance can therefore make overt behaviour unusually uninformative about internal activation. The mechanism is not unique to any nation or culture and must be tested at the relevant organisational or local scale.

Let e_i be the fraction of time in the bottleneck, S_i the degree of expressive suppression, $P_{base,i}$ baseline metabolic power, and $P_{vig,i}$ genuinely measured monitoring expenditure. A proposed effective suppression-power term is:

$$P_{sup,i} = e_i \cdot (\alpha A_i S_i + \beta A_i^2 S_i^2)$$

$$P_{bio,i} = P_{base,i} + P_{vig,i} + P_{sup,i}$$

The parameters α and β are non-negative and have units of watts when A_i, S_i and e_i are dimensionless. These relations are candidate effective-power models, not established universal calorimetry. Sympathetic arousal, cardiovascular response and allostatic burden cannot be converted into watts merely by assumption. Direct metabolic measurement or a validated conversion is necessary to estimate these coefficients [10].

To compare expressive regimes, use $\Delta P_{net,i} = P_{sup,i} - P_{avoided,i}$, where $P_{avoided,i}$ is the measured or estimated cost of conflict, unwanted attention or escalation avoided by suppression in the explicit counterfactual. Thus suppression need not always raise net immediate energy expenditure. The relevant long-run hypothesis concerns repeated unresolved activation and inadequate recovery.

Let L_i be an effective allostatic-burden coordinate within b(t), $Rrec_i$ normalised recovery availability, and k_A, k_S, k_R calibrated rate coefficients. Then:

$$dL_i/dt = k_A e_i A_i + k_S e_i A_i S_i - k_R Rrec_i L_i$$

Accumulation occurs whenever $k_A e_i A_i + k_S e_i A_i S_i > k_R Rrec_i L_i$. This expression allows chronic exposure and restricted recovery to matter without insisting that high visible arousal persists indefinitely. Under comparable material exposures, a high-suppression regime has a larger conditional burden if $k_S > 0$ and the avoided-cost term does not offset it. This is a falsifiable hypothesis; current expressive-suppression experiments support some acute physiological effects, not a calibrated society-level burnout law [10].

A parsimonious behavioural horizon function is:

$$H_{i,d}(t) = min(H_{feasible,i,d}(t), H_{ref,i,d} / (1 + \lambda L_i + \mu V_i))$$

Here $H_{ref,i,d}$ is a reference willingness-to-commit horizon and $H_{feasible,i,d}$ is the independently assessed physically supportable horizon. λ and μ must be fitted. The minimum is important: perceived confidence cannot extend an obligation beyond feasible supply, while feasible supply does not ensure that the actor will accept an unverified obligation. Horizon compression can be an adaptive allocation away from unreliable long-duration commitments, not a direct measure of individual dysfunction or proof that successor cooperation is impossible.

4.1 Neurophysiological boundary

Ensemble Synchronisation and the Dynamical Reality of Somatic Signals (The Section 9.5 Coupling): A critical methodological distinction separates single-subject laboratory psychophysiology from macroscopic observation in shared physical bottlenecks. For any single commuter i observed in isolation, markers such as sweating, muscular stiffness, gaze aversion, repetitive self-touching, or headphone use are subject to individual confounders (thermal load, illness, fatigue, or personal habit). However, as established in Section 9.5 of the core framework (Synchronisation and Macroscopic Signalling), human environmental sensitivity is distributed continuously across b(t). In a well-buffered, predictable niche, individual autonomic fluctuations remain asynchronous and cancel out across a crowd as uncorrelated noise. When niche degradation erodes baseline behavioural predictability in shared public bottlenecks, the high-plasticity vanguard disinhibits first under GUTS and broadcasts multi-modal distress cues (apocrine chemosignals, kinematic tension, startle reactivity, and covert visual scanning) that entrain neighbouring nervous systems.

When an observer repeatedly registers that a large fraction of commuters across multiple carriages and days are simultaneously exhibiting defensive somatic guarding and prosthetic sensory bunkering, uncorrelated individual confounders cancel out by ensemble averaging. What the human observer registers is a **phase-synchronised macroscopic signal** on $t_{autonomic}$.

Methodological Lock on Signal Reality (The Bolting-Horse Principle): Within the coupled state-space $x(t) = (b(t), m(t), a(t), u(t))$, the downstream dynamical effect of a synchronised Layer 1 alarm depends on the **recognition and propagation of the signal within b(t)**, not on whether an external institutional model in $a_{models}(t)$ judges the alarm to be statistically proportionate. In accordance with evolutionary signal-detection theory (Nesse, 2005; Chapter 10), mammalian threat sensors are selected to prioritise false alarms over fatal misses. Once a synchronised somatic unsafety signal is broadcast and mutually recognised across a shared bottleneck, the resulting autonomic load (L_i), expressive suppression cost (S_i), and horizon compression (H_i,d downward) are physically and metabolically real regardless of whether institutional crime surveys classify the fear as "irrational" or "misperceived." A horse bolting in panic at a bumblebee exerts the same physical force as a horse bolting from a predator; similarly, a population synchronised into daily somatic unsafety undergoes real allostatic attrition and behavioural withdrawal from N_0 even while official representations in $a_{models}(t)$ insist the niche remains safe.

4.2 GUTS–ACM coupling and the four-dimensional topology of prosthetic safety enclaving

While Layer 1 autonomic unsafety operates on the fast physiological clock ($t_{autonomic}$) and the Adaptive Calibration Model (ACM; Del Giudice et al., 2011) operates across developmental and intermediate signalling clocks (t_{dev} and t_{sig}), the two mechanisms are coupled in series within b(t):

- **GUTS as the proximal input gate:** Under the Generalized Unsafety Theory of Stress (Brosschot et al., 2016, 2018), neuroendocrine stress readiness is the default state of the organism, requiring continuous, credible cues of environmental and relational safety to maintain prefrontal inhibition. Consequently, the biological sensor does not calibrate primarily to rare, discrete mortality events (such as statistical homicide rates); it calibrates to the continuous presence or absence of social slack, predictable conspecific co-regulation, and controllable territorial refuge. When structural cannibalisation and rate-induced compression ($t_{fin} << t_m < t_{inst}$) hollow out household time buffers, alloparental redundancy (Σ_{care}), and behavioural predictability in shared public spaces, GUTS inhibition fails chronically (driving q_i,d(t) and $A_i(t)$ upward in Section 4).
- **ACM as the developmental and behavioural router:** Chronic GUTS disinhibition generates the sustained allostatic accumulation ($dL_i/dt > 0$ in Section 4) that the ACM switchboard integrates to route phenotypes across t_{dev} and t_{sig}. Where cohorts lacking material buffers or facing externalising selection pressures adapt via second-order physiological blunting (the ACM unemotional / deactivating profile, reducing autonomic receptor sensitivity to survive unavoidable exposure, corresponding to the habituation/occlusion caveat in Section 8.3), high-sensitivity cohorts ("orchids") and cohorts structurally dependent on formal institutional trajectories calibrate disproportionately toward the ACM vigilant-anxious profile.

The Law of Prosthetic Safety Substitution: Because organisms calibrated to the vigilant-anxious profile experience acute metabolic attrition when exposed to unbuffered GUTS disinhibition—compounded by expressive suppression costs $S_i(t)$ in conflict-avoidant cultures (Section 4)—they cannot achieve sufficient physiological recovery R_{rec},i(t) inside the degraded shared niche N_0. When spontaneous, low-cost social safety cues vanish from N_0, actors must

expend private resources, financial leverage, or regulatory effort to construct prosthetic safety enclaves (B_{sub}). This enclaving operates across four structural dimensions of boundary gating:

- **Spatial-territorial gating (the static sanctuary):** The physical home is repurposed from a site of social hospitality into a sealed autonomic decompression chamber ("cave"). To prevent territorial crowding stress during extended indoor retreat, the sanctuary must be sufficiently spacious ("cave, not cage") and tenure-secure to register immediate sovereignty on $t_{autonomic}$—driving households to accept high long-horizon financial leverage on t_{fin} (mortgage debt) in exchange for immediate daily GUTS inhibition and recovery $R_{rec},i(t)$. To protect the recovery sanctuary from horizontal status auditing and social-script friction, non-kin entertaining inside the home is curtailed in favour of neutral commercial third-places, outer doors are gatekept against unplanned neighbour or service-worker contact, and surrounding residential landscapes are stripped of visual occlusion (foliage clearing and floodlighting) to permit immediate visual confirmation of conspecific absence.
- **Kinetic-corridor gating (the mobile capsule):** Because moving between static sanctuaries requires traversing shared public bottlenecks where ensemble-synchronised Layer 1 distress is concentrated (Section 4.1), actors deploy mobile physical prosthetics to drive bottleneck exposure toward zero ($e_i,d(t)$ -> 0 in the Section 4 allostatic equation). In urban transport, this manifests as the substitution of oversized, acoustically isolated private vehicles (functioning as armoured mobile caves) for shared transit, or the deployment of active noise-cancelling audio and visual screen shields when shared transit is unavoidable. Across t_{dev}, high-plasticity juveniles whose nervous systems cannot tolerate the phase-synchronised stress of public transit corridors and under-buffered schools execute biological bottleneck refusal (chronic school non-attendance), or require parental physical escorting as a mobile co-regulatory shield.
- **Virtual-temporal gating (asynchronous digital decoupling):** Digital environments (remote work and study, multiplayer gaming, social media, text messaging, parakin streaming, and digital dating or gambling) function as spatio-temporal safety valves. Unlike unmediated physical encounters—which require real-time autonomic co-regulation and carry unscripted social or physical vulnerability—virtual interfaces decouple social interaction and foraging feedback from both physical co-presence and synchronous time. The screen provides simulated social interaction and predictability with an instantaneous termination switch, allowing the organism to harvest social or dopaminergic cues without exposing its physical perimeter. At the intimate limit, when daily allostatic load $L_i(t)$ leaves minimal margin for $R_{rec},i(t)$, the nervous system increasingly vetoes physical pair-bonding and sexual intimacy inside the home sanctuary to avoid introducing partner unpredictability into its primary recovery zone.
- **Normative-institutional gating (bureaucratic "safe spaces"):** In institutional arenas where physical withdrawal into a private cave is impossible (schools, universities, and corporate workplaces), cohorts calibrated to the ACM vigilant-anxious profile attempt to enforce GUTS inhibition through administrative rule-making in $a_{rules}(t)$. Demands for institutional "safe spaces," speech and interaction codes, and formalised conflict-mediation protocols represent an attempt to eliminate unscripted interpersonal prediction errors by statute—converting deficits in organic Layer 1 and Layer 2 relational reliability into rising bureaucratic verification and policing overhead (P_{vig} in Section 1.2).

Methodological Lock on Causal Direction (Inverting the Digital and Overprotection Fallacies):

Conventional sociological accounts (e.g., Haidt, 2018, 2024) attribute rising youth anxiety, declining independent childhood mobility, and demands for institutional "safe spaces" to

exogenous cultural or technological errors—specifically the adoption of smartphones and social media alongside an irrational cultural shift toward "overprotective parenting" during a period of low statistical violent crime. Within the DST–NCT framework, this attribution reverses cause and effect by conflating statistical crime rates with GUTS safety cues:

- Parental restriction as niche sampling: Independent juvenile exploration requires redundant alloparental monitoring across the local community (the reproductive support network and Σ_{care} in Sections 6.1–6.2) and low-stakes developmental margins. When dual-earner t_{fin} labour compression, kin-network thinning, and hyper-assortative educational sorting eliminated communal slack and turned public transit corridors into unmonitored stress bottlenecks, parents restricted juvenile roaming and escalated escorting because their biological sensors registered the disappearance of external co-regulatory buffers.
- Virtual retreat and "safe spaces" as downstream prosthetics: While algorithmic media amplification can secondarily distort horizontal priors (Section 8) and reduce real-world conflict tolerance, the primary migration of youth and adults into virtual environments, private spatial caves, and bureaucratically policed "safe spaces" is an endogenous response to physical and relational unsafety in N_0. Treating digital screens or parental caution as the root cause of demographic anxiety mistakes the prosthetic analgesic for the underlying niche contraction.

5 Hysteresis, public recognition and transmission topology

The onset of Layer 1 unsafety, pragmatic withdrawal and public acknowledgement need not coincide. Private observation of a discrepancy is not common knowledge. Inherited normality schemas can maintain outward procedural performance after practical reliability has contracted. The key transition mechanism is therefore an explicit threshold for public expression and coordination rather than an assumed instantaneous replacement of developmentally acquired schemas.

Define three transmission components for observer i: Z_i is vertical cumulative exposure to caregiver stress, defensive provisioning and developmental obstruction; O_i is oblique mismatch between institutional declarations and materially costly operator behaviour; K_i is the weighted visible fraction of relevant peers acknowledging or acting upon the same discrepancy. Let θ_i denote the public-disclosure threshold and $Sanc_i$ the cost of public departure from a continuity script. A threshold score is:

$$J_i = \alpha_D D_{ipositive} + \alpha_Z Z_i + \alpha_O O_i + \alpha_H K_i - \theta_i - Sanc_i$$

Public revision is observed when $J_i \geq 0$. Each component must be normalised, and α terms estimated from longitudinal evidence or experimentally elicited threshold choices. Visible private routing alone does not establish operator extraction: oblique CRED divergence concerns credibility, while Chapter 12 fiduciary defection requires evidence of actual redirected institutional resources [11, 12].

For a tractable network approximation let Q be the population fraction whose public expressions have changed. The update $Q(t + \Delta t) = F(\alpha_0 + \alpha_H Q(t))$ represents heterogeneous disclosure thresholds. A local amplification condition is $\alpha_H F'(\alpha_0 + \alpha_H Q) > 1$. A finite discontinuity additionally requires multiple fixed points and a disturbance large enough to cross the intervening unstable branch. This is not a universal claim that all growing discrepancies lead to abrupt transition. Fragmented horizontal networks can sustain multiple local public-recognition states without a national snap [13, 14].

Vertical transmission can progressively change cohort-specific θ_i without generating an immediate public event. Oblique CRED divergence can provide an unusually legible focal observation but cannot alone establish population-wide recognition. Horizontal ties transmit the observation and make others' knowledge partially visible; their topology is decisive for collective thresholds. In highly segmented networks, abrupt shifts remain local. A care-boundary breach can operate as a focal event when independently experienced and mutually known, but it is neither a necessary nor sufficient cause of a nation-wide Phase III.

Hysteresis is explicitly path-dependent. Let d_{on} be the divergence threshold required to enter a new public coordination branch, and d_{off} the threshold below which the former branch is regained on the same divergence scale. In an hysteretic regime $d_{off} < d_{on}$. Restoring old survey scores or public reassurances does not prove restoration of lower-layer reliance. Conversely, public discourse can switch while internal declarative schemas remain temporally stable. Finally, a public-recognition snap is an informational and behavioural event; it must never be equated automatically with crossing the physical separatrix Σ between N_0 and N_i.

6 Reproductive continuity, care boundaries and coercive command

6.1 Three distinct Chapter 10 networks

The reproductive support network comprises actual care, provisioning and access to physical resources that support development and subsequent household formation. The recognised succession field identifies the younger cohorts whose continuity is salient to a given observer. The trusted observation network conveys testimony and shared visibility concerning that field [1]. These networks overlap but are not coextensive. Layer 1 samples local bodily safety and care; Layer 2 observes reliable realised support and contingent fallback; Layer 3 encodes historical expectations about which succession cues signify continuity.

The Parakin Placebo is a testable divergence between symbolic continuity and real reciprocal provisioning. Let C_i,succ be declared or perceived succession continuity, $M_{i,para}$ mediated parasocial succession exposure and $R_{i,care}$ independently observed care reciprocity. A parakin substitution is identified when increases in $M_{i,para}$ sustain $C_{i,succ}$ without corresponding increases in $R_{i,care}$ or materially realised intergenerational support. This is a signal substitution, not material replacement. Digital channels that mobilise real care, labour and resources instead constitute part of the actual support network; calling all mediated support a placebo would erase this distinction.

6.2 Σ_{care} and the physical-recognition gap

The care boundary has at least two distinct aspects: the physical inability of a specified provisioning arrangement to supply care on time and the public coordination transition produced when its limitations become widely recognised. Local physical care deterioration can precede general public acknowledgement. A hardened test therefore follows the full needs-defined cohort, including people refused entry to an administrative system, and separates care required, care actually delivered, care transferred to households and requirements remaining unmet. Re-familialisation transfers a physical workload to the remaining biological network; it does not eliminate that workload [1].

Continued reliance on formal care under no feasible alternative is not proof of Layer 2 trust. Withdrawal from incumbent care does not prove disappearance of total care provision: households, local organisations and successor networks can supply care subject to their own

labour, material, timing and recovery constraints. The viability of N0 and Ni is assessed separately.

6.3 Chapter 13: coercive predictability versus relational reliance

A force or emergency hierarchy can be predictably obedient under surveillance and sanction while its operators do not trust higher echelons to deliver reliable logistics, casualty evacuation or truthful reporting. Coercive predictability belongs to $a_{rules}(t)$, whereas Layer 2 relational trust is the continued willingness to expose personnel, materiel or dependants to unverified counterpart commitments. The two need different operators and observations.

Define $Q_{required}$ the task-specific necessary supply throughput and $Q_{effective}$ the supply delivered on time. A useful bottleneck approximation is $Q_{effective} = \min(Q_{transport}, Q_{supply}, Q_{handling}, Q_{coordination})$. The components share the appropriate units of delivered flow. Verification latency t_{verify} should be compared with available decision time $t_{decision}$. When $t_{decision} < t_{verify}$, repeated local checking can make accurate response physically too slow. Under $T_{recur} < T_{refill}$, reserve depletion further tightens the trade-off between checking, movement, repair and care. Increased coercive enforcement can sustain outward compliance while deteriorating truthful negative reporting and the accuracy of $a_{models}(t)$. It does not follow that strict hierarchy always reduces cohesion; reliable provisioning and predictable rules can preserve coordination under threat [1].

The diagnostic observation is therefore not the number of orders obeyed, but delivery reliability, independently verified local stock, report accuracy, casualty-care performance, latency and function after command-node loss. A lower-overhead modular command arrangement counts as Ni only if its own material and operational continuities persist, not merely because the original command chain weakened.

7 Goodhart-resistant five-domain telemetry matrix

No proxy is absolutely non-gameable. An externally instrumented signal may be altered by selection of observation sites, exposure withdrawal, changed case mix or diversion of resources to sampled units. A physically realised outcome is more difficult to relabel but still requires fixed denominators, independent verification and accounting for displaced costs. The model therefore grades proxies by manipulation channels rather than declaring them non-reactive [15].

Table 2. Hardened telemetry matrix

Domain	Primary observation	Principal gaming or confounder	Hardened specification
Shared physical bottlenecks	Repeated within-person autonomic deviation and avoided exposure	Screens, headphones, sweating and movement are non-specific; selective sampling	Match noise, crowding, temperature, health and trip purpose; record exposure, recovery and independently verified hazards
Workplace coordination	Verification and rework per completed physical or service task	Renaming paperwork; off-ledger checking; changed task mix	Count all paid and unpaid verification minutes, handovers, errors and completion time under stable task definitions
Care interfaces	Delivered versus need-adjusted care within deadline	Eligibility rules, waiting-list exclusions and case reclassification	Independent needs sampling including rejected cases; measure delivery, family transfer and unmet care
Operator behaviour	Divergence between public commitments, exposure and controlled resource routing	Lifestyle-based attribution; hidden beneficial ownership; staged appearances	Audit procurement, ownership, allocation authority and actual resource flows; separate private hedging from defection
Survey–behaviour difference	Within-domain difference between stated confidence and realised unverified reliance	Question wording, cohort replacement and compositional change	Panel observation on same counterpart, care domain, exposure and horizon; control options and physical outcomes

A high-protection proxy is one for which changing the reported value without changing delivered physical capacity requires substantial, independently detectable reallocation. In practical ranking for physical-state inference, independently verified care delivery is strongest; completed-task organisational friction and traced operator allocations are intermediate; non-specific public somatic displays and cross-domain survey gaps are weaker. For early warning the order can differ. The ranking is not a universal statement about social importance or a numerical likelihood of failure.

A non-circular inference protocol requires independent reconstruction of the physical variables before trust observations are used to explain or predict their movement. Trust signals may improve the estimate of a hidden state by supplying additional observations; they cannot constitute both the sole evidence of structural decline and its explanation.

8 Boundary conditions and observational edge cases

8.1 Acute exogenous shock versus endogenous contraction

A shock in u(t) can produce a rapid Layer 1 response, temporary Layer 2 verification and a comparatively unchanged Layer 3 schema while underlying buffers and physical services recover. By contrast, independently established material decline combined with repeated shortfalls predicts persistent lower-layer recalibration, incomplete recovery and broader diffusion across domains. A severe shock may reveal or accelerate an existing endogenous constraint. The discriminator is the time relation between measured physical restoration, repeat exposures and behavioural recovery, not the mere presence of defensive signalling.

8.2 Digital mediation of horizontal sensing

Let W_{ij} describe an underlying observation tie, p_{ij} the probability that a platform delivers an observation, and c_{ij} its credibility weighting. Effective mediated exposure is $W_{effective,ij} = W_{ij} \cdot p_{ij} \cdot c_{ij}$. Platforms can amplify one event repeatedly, suppress observation of widespread physical shortages or split populations into separate locally coherent recognition networks. Multiple impressions of a duplicated claim are not multiple independent pieces of evidence. A digital parakin cue can sustain Layer 3 continuity without a real support tie, but digital coordination that mobilises real resources is part of b(t), m(t) and a(t), not a placebo [16].

8.3 Occlusion, trapped participation and reverse selection

Repeated unsafety can yield habituation, fatigue or altered response profiles rather than indefinitely rising visible arousal. Actors unable to exit may continue formal participation despite low Layer 2 reliance. Exit and cohort sorting can leave an administratively observed population with higher apparent trust while the original cohort's willingness to rely has fallen. Longitudinal observation of entrants, leavers and rejected applicants is therefore indispensable. Declining N0 macro-edge trust can coexist with rising dense internal Ni cooperation. Mean trust does not identify the viability of either regime.

9 Comparative three-configuration test at t_0 = 2026

The country labels below identify ideal-typical institutional configurations, not homogeneous national populations. Observed numerical indicators are kept distinct from predicted unmeasured three-clock values. Data years differ and no current cross-country dataset combines wearable autonomic sampling, observed domain-specific reliance, longitudinal schemas and physical buffer balances. Accordingly the phase portraits are provisional qualitative positions, not calibrated state estimates. The empirical comparison tests whether the model generates different conditional observations and falsifiers beyond surveys and GDP.

9.1 Sweden — high-schema-inertia configuration

The working configuration proposes elevated inherited interpersonal trust and comparatively stable declarative continuity alongside heterogeneous practical guarding around urban interfaces and care access. Integrated Values Surveys data around 2022 place Sweden at approximately 63% responding that most people can be trusted; the OECD's different 2025 instrument reports 42% high or moderately high trust in national government [17, 18]. These values should not be combined as a continuous Layer 3 series.

Materially relevant observations are more specific. The Swedish Crime Survey reports 25% feeling unsafe or avoiding their residential area outdoors late at night in 2025, essentially unchanged from 2024 and lower than 28% in 2021; this does not support a claim of uninterrupted nationwide deterioration [19]. School segregation has stabilised at a higher level than twenty years earlier and reflects both residential patterns and school choice, but does not uniquely reveal distrust [20]. A March 2026 assessment of eldercare records roughly double labour-market-average sickness absence among assistant nurses and significant recruitment needs through 2030, indicating stress on a particular care labour system without proving Σ_{care} has been crossed [21].

Conditional phase portrait: Layer 3 remains comparatively high in interpersonal domains; Layers 1 and 2 have wide exposure- and service-specific dispersion. Where actual care and organisational reliability deteriorate, $D_{ipositive}$ widens before national declarative measures necessarily move. The most plausible near-term observational basin given current evidence is persistent cross-domain heterogeneity with partial local divergence, followed by segmented public-recognition events if repeated physical shortfalls become independently visible. A nation-wide Phase III snap ranks below these alternatives absent evidence of broad shared thresholds and cross-network propagation.

Discriminating test: Follow the same commuter, employee and eldercare-recipient panels. Estimate exposed versus unexposed autonomic recovery, verification person-hours and needs-adjusted care delivery, while measuring the same domain-specific declarative expectations. If practical trust and physiological safety remain aligned with declared confidence after exposure

and resource access are controlled, the proposed hidden divergence is weakened. A population mean alone does not test the double-bind.

9.2 United Kingdom — externally coupled import and claims configuration

The working configuration combines unequal private fallback buffers, lower political confidence and physically material external dependence. The Integrated Values Surveys comparison gives approximately 43% generalised interpersonal trust around 2022, while the OECD's 2025 government-trust measure reports 33% [17, 22]. The ONS reports 29.2% tending to trust the UK government in March 2026 under its distinct instrument; the same source records substantial financial difficulty within the adult population [23]. ONS household-wealth statistics for April 2020–March 2022 document marked wealth heterogeneity, a constraint on equal access to alternative buffers rather than an immediate measurement of contemporary trust [24].

The UK's official 2025 energy digest records net import dependency of 43.8% in 2024. NHS England's 2024/25 estate returns report an estimated £15.9 billion cost to eradicate backlog maintenance, an accounting estimate of restoration expenditure rather than a direct exergy stock [25, 26]. These observations motivate testing distinct domestic and external interfaces; they do not jointly prove a national physical decline trajectory.

The framework's Chapter 6.9 specifies that shipments require physical transport, claims settlement, legal and contractual coordination and information. Accordingly one must observe both $T_{pragmatic}$(household → domestic care or infrastructure operator | d,H,t) and $T_{pragmatic}$(external counterparty → UK importer or clearing node | d,$H_{ext,t}$). The latter is not the domestic citizen's survey sentiment. Nor does an external party's positive forward horizon $H_{ext} > 0$ measure reserves or guaranteed delivery. It represents an expectation that must be tested against actual settlement, credit, insurance, shipping and physical availability.

Conditional phase portrait: The salient feature is not a particularly large Layer 3 lag, but pronounced domain, class and counterparty segmentation of Layer 2 and an external-reliance axis absent from a national trust scalar. Conditional on continuing foreign delivery and settlement, the most accessible configuration is differentiated domestic substitution with sustained cross-border flows. The next-ranked risk configuration is an external coordination interruption coincident with low domestic structural redundancy. Abrupt loss of all external confidence is not established by current data and ranks below these more limited dependencies.

Discriminating test: Observe changes in supplier credit terms, insurance acceptance, real goods arrival times, port bottlenecks, repair lead times and domestic alternative capacity against fixed needs-defined cohorts. If external contractual confidence declines but delivered physical throughput is maintained through substitutions, the predicted high dependency of effective service on that specific counterparty relation is weakened. If domestic trust changes without material or contractual correlates, the inference to structural cannibalisation is likewise weakened.

9.3 United States — distributed modular-buffer configuration

The US configuration is marked by strong internal variation in institutional confidence, information channels and material provisioning. Integrated Values Surveys data give approximately 37% generalised interpersonal trust around 2022 [17]. In September 2025, Pew found that 17% trusted the federal government to do what is right almost always or most of the time; responses were strongly partisan, so the average conceals opposed political evaluations rather than a single national schema [27]. Pew's 2025 news research independently identifies strongly differentiated

source reliance across partisan audiences [28]. Neither measure observes Layer 1 autonomic safety.

The material case for modular buffering is more specific. A Congressional Research Service review catalogues announced data-centre electricity arrangements involving dedicated on-site natural-gas generation, additional transmission capacity and long-term nuclear procurement. Some projects remain proposals and contracted access to generation is not the same as an independent energy island [29]. A dedicated baseload contract can represent rational continuity investment, bargaining power or commercial risk management; it does not itself prove corruption or fiduciary capture. $N_{i,capture}$ should be reserved for demonstrated privileged reallocation of common buffers combined with an operationally distinct and potentially persistent successor configuration.

Conditional phase portrait: The US is better represented as a distribution or set of local basins than as a single point in (A,V,C). Some organisations and regions retain low-verification cooperation, while others depend on contractual enforcement, protected infrastructure or bounded alliances. The most accessible observation regime is asynchronous reallocation and multiple local schemas; selective corporate infrastructure provisioning follows where physical demand and finance permit it. A synchronous national Phase III ranks below modular divergence absent cross-coalition common recognition. The possibility of genuine infrastructural expansion in some nodes remains open and must not be misclassified as general cannibalisation.

Discriminating test: Compare grid-access and outage durations, dedicated generation actually commissioned, contractual versus physically delivered energy, employee and contractor verification friction, emergency care and dependency chains across matched regions. If these measures show convergence despite polarised declarative views, the hypothesis that information fragmentation maps onto separate practical cooperation basins is weakened. If privately buffered nodes merely shift fuel, maintenance or protection costs to public systems, their independence as Ni must be rejected.

9.4 Comparative discrimination and basin accessibility

Table 3. Conditional phase portraits, 2026

Feature	Sweden	United Kingdom	United States
Declared Layer 3 configuration	Relatively persistent interpersonal normality	Lower political confidence; domain-specific residual deference	Strongly coalition- and institution-dependent
Layer 1 hypothesis	Exposure-specific unsafety and suppression; not nationally calibrated	Service-, location- and resource-access differentiation	Region-, occupation- and infrastructure-specific variation
Layer 2 hypothesis	Quiet auditing and substitute care where reliability falls	Asset-stratified fallback; domestic and foreign counterpart reliance	Dense bounded networks and selective dedicated physical buffering
Principal hidden relation	C_i can remain above local R_i	External settlement and physical delivery can diverge from domestic surveys	Network modularity and actual material independence differ from national means
Most accessible conditional path	Local divergence, then segmented recognition	Stratified fallback while foreign flows continue	Asynchronous multi-basin reallocation
Less accessible without further evidence	Nation-wide sudden schema snap	Generalised foreign counterparty flight	Synchronous nationwide public snap
Key falsifier	No within-domain C_i–R_i divergence after controls	No delivery or capacity consequences of assumed interface weakness	Polarised surveys without distinct realised cooperation or buffers

The ranking concerns relative accessibility under the stipulated configurations, not probabilities assigned to whole countries. Sweden's potential primary observability failure is **concealed local divergence**; the UK's is **misidentification of domestic versus international coordination capacity**; the US's is **aggregation across simultaneous but non-equivalent regional and coalition basins**. Existing OECD/WVS instruments record stated responses and standard macro accounts record monetary or output aggregates. Neither independently measures the domain-specific vulnerable commitment, energetic verification cost, care requirement actually fulfilled, counterparty delivery latency or cross-network threshold necessary to discriminate these mechanisms.

9.5 Comparison cannot establish unmeasured parameters

None of the three phase portraits supplies estimates of α, β, k_S, thresholds θ_i or actual national A_i distributions. The three-country exercise therefore establishes discriminating hypotheses and an observation design, not parameter identification. Reported national survey figures come from instruments with different wording and sampling periods and must not be plotted as a calibrated common-scale time series. A future empirical test needs harmonised questions, exposure-adjusted repeated measurement, physically grounded service outputs and pre-specified network samples.

10 Identification strategy and testable predictions

A minimum empirical panel combines (i) fixed-denominator needs or task observations, (ii) repeated within-person or within-unit somatic and behavioural measurements, (iii) directional declarative questions referring to the same counterparty, domain and horizon, (iv) independent material and care-buffer audits, (v) mapped information and support networks, and (vi) observation of actor exit, exclusion and successor routing. Data should be sampled before and after identifiable disturbances wherever possible. Quasi-experimental events include sudden failures followed by independently verified repair, exogenous supply interruptions, changes in reporting requirements without physical capacity change, and gradual degradation across otherwise comparable local units.

Prediction P1: Under independently verified material service deterioration, practical checking and reliance shift before declarative continuity in cohorts with greater inherited schema inertia, after adjusting for alternatives and exposure. A failure of this ordered lag weakens the three-clock diagnostic for that population.

Prediction P2: Under comparable unsafety exposure and reduced recovery, high expressive suppression is associated with larger cumulative physiological burden and earlier practical horizon compression than in matched low-suppression conditions. The prediction must be tested with actual physiological recovery and task behaviour, not visual impressions alone.

Prediction P3: Where network reinforcement and threshold heterogeneity generate multiple stable public-expression branches, an increase in cross-network common visibility can produce abrupt public revision without a simultaneous discontinuity in B(t) or s(t). Absence of such network effects under adequate sampling weakens the proposed hysteresis mechanism.

Prediction P4: Administrative changes that improve reported compliance without improving fixed-denominator delivered care, repair or physical throughput will increase the observed divergence between $a_{models}(t)$ and independently audited outcomes. If real capacity rises proportionately, an interpretation of gaming is unsupported.

Prediction P5: Declining reliance across N0 macro-edges can coexist with increasing internal cooperation in Ni. The structural question is whether the alternative network actually delivers sustained care, material inputs and replenishment; increased in-group sentiment alone is insufficient.

Prediction P6: For import-dependent configurations, the explanatory power of external counterpart reliability for timely physical delivery remains after controlling for domestic declarative trust, provided the physical route and supply remain available. If contractual confidence and actual delivery are decoupled through other routes, the corresponding observed interface must be revised.

A critical observation-to-state restriction applies throughout: the same physiological or survey response may arise from several x(t) configurations, and an institutional intervention can change visibility without changing substrate. The model's claimed gain is improved discrimination using multiple non-equivalent observations, not universal observability from trust alone.

11 Discussion and regime-neutral implications

The model's main extension is the separation of biological telemetry, practical reliance, declarative continuity and physically delivered buffering across different response times. These layers can interact, but their measurement dimensions and causal roles cannot be collapsed. A policy, managerial or rhetorical campaign that increases stated confidence can conserve verification expenditure only if it changes actual coordination behaviour, and that change cannot be counted twice as new exergy or structural capacity. Conversely, biological detachment from N0 need not represent loss of wider human organisation. It may generate viable alternative networks, restricted fallback systems or configurations that fail their own care and material constraints. Assessment is regime-specific rather than incumbent-relative.

The strongest caution concerns overreach. Physiology is a noisy indicator of exposure, not a direct detector of institutional decline. School or residential sorting does not automatically reveal untrustworthiness or resource capture. Public confidence is not a homogeneous national characteristic, and survey differences cannot identify a single historical stage. The stylised

comparison supports the distinctive questions the telemetry model asks; the empirical establishment of all three clocks at national scale remains outstanding.

The proper scope is an observational interface to DST–NCT. Under physical contraction the sensor can reveal where organisms and organisations are recalibrating before administrative models acknowledge the altered trajectory. Under material expansion it can identify distributional, network or informational divergence without misclassifying it as an energy crisis. Under a regime transition it can distinguish the disappearance of an incumbent coordination architecture from the persistence or reconstruction of viable human support networks.

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